

AP Statistics FRQ Practice Exam 12

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AP Statistics FRQ Practice Exam 12

APSTAT-FRQ12-001

Section: FRQ

Type: frq

Primary Unit: Unit 1, Exploring One-Variable Data and Collecting Data

Secondary Unit: Unit 4, Inference for Quantitative Data: Means

FRQ Type: Formulate Questions and Collect Data

Prompt:

A bakery wants to compare two compostable wraps, A and B, for limiting moisture loss from bread. From the 300 loaves produced in one morning batch, the bakery takes a simple random sample of 30 loaves. Each selected loaf is cut lengthwise into two comparable halves. For each loaf, one half is wrapped with A and the other with B. After 48 hours under standardized storage conditions, the response is the percentage of the half-loaf's initial mass that has been lost.

The results are summarized below.

Wrap	Mean percentage mass loss
A	6.4%
B	8.1%

For each loaf, define d = mass-loss percentage for its A half – mass-loss percentage for its B half. The mean of the 30 paired differences is -1.7 percentage points, with a standard deviation of 2.2 percentage points.

- (a) Identify the population, sample, experimental units, blocks, treatments, and response variable.
- (b) State a statistical question that the experiment can answer, and define an appropriate population parameter.
- (c) Describe how the two wraps should be randomly assigned within each loaf. Identify one important condition, other than wrap type, that should be controlled.
- (d) Calculate and interpret the observed mean paired difference. Explain one statistical advantage of pairing the two halves from the same loaf.
- (e) Describe the appropriate cause-and-effect and population conclusions. Identify one limitation on extending the result to other settings.

Scoring Guide:

Total Points: 10

(a) 2 points

- Point 1: Identifies the population as the 300 loaves in the morning batch and the sample as the 30 selected loaves.
- Point 2: Identifies the 60 half-loaves as experimental units, the 30 original loaves as blocks or matched pairs, wraps A and B as treatments, and percentage mass loss after 48 hours as the response.

(b) 2 points

- Point 1: States a comparative question about whether wraps A and B differ in mean 48-hour percentage mass loss for loaves in the batch under the specified storage conditions.
- Point 2: Defines μ_d as the population mean of A-minus-B paired mass-loss differences for the 300 loaves in the morning batch under those conditions.

(c) 2 points

- Point 1: Describes an independent chance process within each loaf, such as a coin flip, to assign A to one half and B to the other, with the assignment repeated separately for all 30 loaves.
- Point 2: Identifies a relevant controlled condition such as half size, storage temperature, humidity, storage duration, initial weighing procedure, or shelf position.

(d) 2 points

- Point 1: Calculates or reports $6.4 - 8.1 = -1.7$ percentage points and interprets this as wrap A producing 1.7 percentage points less mass loss on average for the sampled loaves.
- Point 2: Explains that pairing controls loaf-to-loaf differences in ingredients, size, crust, or initial moisture, which can reduce unexplained variability in the wrap comparison.

(e) 2 points

- Point 1: Explains that random assignment supports a causal conclusion about the wraps and the original SRS supports generalization to the 300-loaf morning batch.
- Point 2: Limits extension beyond the sampled batch or standardized 48-hour conditions because other recipes, production days, temperatures, humidities, or storage durations may respond differently.

Model Guidance:

The treatments are applied to half-loaves, while each original loaf forms a matched pair. A separate random assignment within every loaf prevents systematic differences between left and right halves from being confused with wrap type. Storage and measurement conditions should remain the same for both halves.

The negative A-minus-B mean difference indicates less average mass loss with A. Pairing makes the comparison more precise by removing much of the natural variation between loaves. Random assignment justifies a causal wrap comparison, whereas the random sample determines the population to which the finding can reasonably be generalized—the single morning batch, not every type of bread or storage environment.

Expected Points:

10

APSTAT-FRQ12-002

Section: FRQ

Type: frq

Primary Unit: Unit 2, Probability, Random Variables, and Probability Distributions

Secondary Unit: None

FRQ Type: Analyze Data, One- or Two-Variable Data and Probability

Prompt:

Three incoming translation requests are independently routed to one of four on-call editors. Each request is equally likely to be routed to any editor. Let X be the number of distinct editors who receive at least one of the three requests.

(a) Explain why there are $4^3 = 64$ equally likely ordered routing outcomes. State all possible values of X , and determine how many outcomes produce each value.

(b) Construct the probability distribution of X . Sketch a probability bar graph with labeled axes and correctly scaled bar heights.

(c) Calculate $P(X \geq 2)$ and $P(X = 3 \mid X \geq 2)$. Interpret the conditional probability in context.

(d) Calculate the mean and standard deviation of X . Interpret both values in context.

(e) Describe a simulation using random digits or a random-number generator to estimate $P(X = 3)$. Include how one trial is performed and how the estimate is obtained from many trials.

Scoring Guide:

Total Points: 10

(a) 2 points

- Point 1: Explains that each of three ordered requests has four independent routing choices, giving $4 \times 4 \times 4 = 64$ equally likely outcomes, and states X can equal 1, 2, or 3.
- Point 2: Counts 4 outcomes with $X = 1$, 36 outcomes with $X = 2$, and 24 outcomes with $X = 3$; acceptable reasoning includes choosing editors and counting onto assignments.

(b) 2 points

- Point 1: Gives $P(X = 1) = 4/64 = 0.0625$, $P(X = 2) = 36/64 = 0.5625$, and $P(X = 3) = 24/64 = 0.375$, with probabilities summing to 1.
- Point 2: Sketches a probability bar graph with X on the horizontal axis, probability on the vertical axis, and bars at 1, 2, and 3 with heights 0.0625, 0.5625, and 0.375.

(c) 2 points

- Point 1: Calculates $P(X \geq 2) = 60/64 = 0.9375$.
- Point 2: Calculates $P(X = 3 \mid X \geq 2) = (24/64)/(60/64) = 24/60 = 0.40$ and interprets it as the probability that all three requests go to different editors, given that at least two editors receive requests.

(d) 2 points

- Point 1: Calculates $\mu_X = 1(0.0625) + 2(0.5625) + 3(0.375) = 2.3125$ editors and interprets it as the long-run average number of distinct editors receiving requests per group of three.

· Point 2: Calculates $\sigma_X = \sqrt{\{1^2(0.0625) + 2^2(0.5625) + 3^2(0.375)\} - 2.3125^2} \approx 0.583$ editor and interprets it as the typical distance of X from its mean over many groups of three requests.

(e) 2 points

· Point 1: Labels editors 1 through 4 and, for one trial, generates three independent equally likely integers from 1 through 4, allowing repeats, then records success if all three integers are distinct.

· Point 2: Repeats the trial many times and estimates $P(X = 3)$ by the number of successful trials divided by the total number of trials.

Model Guidance:

Treat the requests as ordered. There are four choices for each request, so the sample space has 64 equally likely outcomes. If $X = 1$, all three requests go to one editor, giving 4 outcomes. For $X = 3$, the requests go to three different editors, giving $4 \times 3 \times 2 = 24$ outcomes. The remaining 36 outcomes have $X = 2$.

The resulting distribution has most of its mass at $X = 2$. Its mean, 2.3125, is a long-run average and need not be a possible single outcome. The conditional probability in part (c) uses only the 60 outcomes for which at least two editors receive work. A valid simulation must allow repeated editor labels because one editor can receive multiple requests.

Expected Points:

10

APSTAT-FRQ12-003

Section: FRQ

Type: frq

Primary Unit: Unit 4, Inference for Quantitative Data: Means

Secondary Unit: Unit 1, Exploring One-Variable Data and Collecting Data

FRQ Type: Analyze Data, Statistical Inference (Confidence Interval or Significance Test)

Prompt:

A manufacturer produces a lot of 800 child-resistant medicine bottles. A design guideline states that the population mean torque required to open the bottles should not exceed 12.0 pound-inches. An engineer takes a simple random sample of 28 bottles from the lot and obtains

$n = 28$

$\bar{x} = 12.8$ pound-inches

$s = 2.1$ pound-inches

A dotplot of the sample torque measurements is roughly symmetric with no apparent outliers. At the $\alpha = 0.05$ significance level, test whether the population mean opening torque for the lot exceeds 12.0 pound-inches.

(a) Define the parameter of interest and state the null and alternative hypotheses.

(b) Identify the appropriate inference procedure and check all conditions.

(c) Calculate the standard error, test statistic, degrees of freedom, and p-value.

(d) State a conclusion in context using $\alpha = 0.05$.

(e) Describe a Type II error in context and explain one possible consequence of that error.

Scoring Guide:

Total Points: 10

(a) 2 points

- Point 1: Defines μ as the true mean opening torque, in pound-inches, for all 800 bottles in the production lot.

- Point 2: States $H_0: \mu = 12.0$ pound-inches and $H_a: \mu > 12.0$ pound-inches.

(b) 3 points

- Point 1: Selects a one-sample t test for a population mean because torque is quantitative and the population standard deviation is unknown.

- Point 2: Cites the SRS and verifies the 10% condition because $28 \leq 0.10(800) = 80$, supporting independence.

- Point 3: Uses the roughly symmetric, outlier-free dotplot to justify the t procedure for the sample of 28.

(c) 3 points

- Point 1: Calculates $SE = s/\sqrt{n} = 2.1/\sqrt{28} \approx 0.3969$ pound-inch.

- Point 2: Calculates $t = (12.8 - 12.0)/0.3969 \approx 2.02$ with $df = 28 - 1 = 27$.

- Point 3: Calculates the upper-tail p-value $P(T_{27} \geq 2.02) \approx 0.0269$.

(d) 1 point

- Point 1: Because $0.0269 < 0.05$, rejects H_0 and concludes that there is convincing evidence that the true mean opening torque for the lot exceeds 12.0 pound-inches.

(e) 1 point

- Point 1: Describes a Type II error as failing to conclude that the lot's mean opening torque exceeds 12.0 when it actually does, with a possible consequence such as releasing bottles that are more difficult to open than the guideline permits.

Model Guidance:

The parameter is a population mean, and σ is unknown, so a one-sample t test is appropriate. The random sample, small sampling fraction, and shape information support the procedure. The sample mean is 0.8 pound-inch above the null value; dividing by the standard error gives $t \approx 2.02$ and a one-sided p-value near 0.0269.

At the 5% significance level, the evidence supports the conclusion that the lot's true mean exceeds the guideline value. A Type II error goes in the opposite direction from the conclusion reached in this sample: it would occur in another application of the procedure if a genuinely excessive population mean were not detected.

Expected Points:

10

APSTAT-FRQ12-004

Section: FRQ

Type: frq

Primary Unit: Unit 5, Regression Analysis

Secondary Unit: Unit 1, Exploring One-Variable Data and Collecting Data

FRQ Type: Interpret Results, Integrated Analysis and Justification

Prompt:

A cold-chain laboratory compares two insulating liners for identical shipping containers. The laboratory uses 48 containers. It randomly assigns 24 containers to Liner A and 24 to Liner B. Within each liner group, four containers are randomly assigned to each of six controlled ambient temperatures: 15, 19, 23, 27, 31, and 35 degrees Celsius. After six hours, the response y is the mass of ice that has melted, in grams; lower values are better. For each liner, x denotes ambient temperature in degrees Celsius.

The separate least-squares models are shown below. The scatterplots for both liners are roughly linear over the experimental range.

Liner	Least-squares model	r	r^2
A	$\hat{y} = 18 + 2.4x$	0.940	0.884
B	$\hat{y} = 42 + 1.6x$	0.880	0.774

- (a) On one set of axes covering 15°C through 35°C, sketch and label both fitted lines. Calculate the temperature and predicted melted-ice mass at which the two lines intersect.
- (b) Interpret both slopes in context. Explain what the difference between the slopes indicates about the liners' sensitivity to ambient temperature.
- (c) Calculate the predicted melted-ice mass for each liner at 20°C and at 35°C. Identify which liner the models favor at each temperature.
- (d) A technician says, "Liner B is always better because its regression slope is smaller." Evaluate the claim over the experimental temperature range using statistical evidence.
- (e) Explain what causal conclusion the randomized experiment supports, why the fitted values are not guarantees for individual containers, and one limitation on generalizing the results.

Scoring Guide:

Total Points: 10

(a) 2 points

- Point 1: Sketches two increasing lines on common labeled axes, with A below B at lower temperatures, the lines crossing within the displayed range, and B below A at higher temperatures.
- Point 2: Solves $18 + 2.4x = 42 + 1.6x$ to obtain $x = 30^\circ\text{C}$ and substitutes to obtain $\hat{y} = 90$ grams at the intersection.

(b) 2 points

- Point 1: Interprets the slopes: each additional 1°C is associated with predicted increases of 2.4 grams of melted ice for A and 1.6 grams for B, on average, over 15°C to 35°C .
- Point 2: Explains that A's predicted melt increases 0.8 gram more per additional degree, so A is more sensitive to increases in ambient temperature according to the fitted models.

(c) 2 points

- Point 1: At 20°C , calculates A: $18 + 2.4(20) = 66$ grams and B: $42 + 1.6(20) = 74$ grams, so the models favor A by 8 grams.
- Point 2: At 35°C , calculates A: $18 + 2.4(35) = 102$ grams and B: $42 + 1.6(35) = 98$ grams, so the models favor B by 4 grams.

(d) 2 points

- Point 1: Rejects the “always better” claim because a smaller slope describes slower change with temperature but does not by itself determine which fitted response is lower at a particular temperature.
- Point 2: Uses the 30°C intersection to conclude that the models favor A below 30°C , predict equality at 30°C , and favor B above 30°C within the 15°C -to- 35°C experimental range.

(e) 2 points

- Point 1: Explains that random assignment of liner and ambient temperature supports causal conclusions about their effects and their differing temperature responses for the tested containers under the laboratory protocol, while scatter around each line means an individual response need not equal its prediction.
- Point 2: Limits generalization because the 48 containers were not described as a random sample from a broader population and because other container designs, storage durations, contents, or temperatures outside 15°C to 35°C were not studied.

Model Guidance:

The two fitted lines have different intercepts and slopes, so neither liner is predicted to be best at every temperature. Setting the equations equal gives a crossover at 30°C and 90 grams. At 20°C , A has the lower predicted melt; at 35°C , B does. The smaller slope for B means its response changes less rapidly as ambient temperature rises, not that its predicted response is always lower.

Because both liner type and temperature were randomly assigned, differences in the response can be attributed to those treatments for the experimental containers under the laboratory conditions. The lines still describe mean predictions with variation around them. The absence of random sampling and the limited treatment range restrict broader generalization and make extrapolation inappropriate.

Expected Points:

10

SERIES DO-NOT-REPEAT BANK UPDATE, APSTAT-FRQ12

- APSTAT-FRQ12-001: Unit 1 primary; Unit 4 secondary; SRS of 30 of 300 morning-batch bread loaves split into matched half-loaves, with wraps A and B randomly assigned within loaf and 48-hour percentage mass loss as the response; identify halves as units and loaves as blocks, control storage conditions, interpret mean paired difference $A - B = -1.7$ percentage points, and distinguish causal wrap comparison from batch-limited generalization; targets naming loaves rather than halves as treatment units and overlooking the precision benefit of matching.
- APSTAT-FRQ12-002: Unit 2 primary; three independent translation requests routed uniformly among four editors, with X equal to the number of distinct editors receiving work; count 64 ordered outcomes

and obtain distribution $P(1) = 0.0625$, $P(2) = 0.5625$, $P(3) = 0.375$, conditional probability 0.40, mean 2.3125, $SD \approx 0.583$, probability bar graph, and repeated-label simulation; targets treating requests as unordered and forbidding repeated editor labels in the simulation.

· APSTAT-FRQ12-003: Unit 4 primary; Unit 1 secondary; upper-tailed one-sample t test for mean opening torque in an 800-bottle lot using an SRS of 28, $\bar{x} = 12.8$ pound-inches, $s = 2.1$, and a symmetric outlier-free dotplot; test $H_0: \mu = 12$ versus $H_a: \mu > 12$, calculate $SE \approx 0.3969$, $t \approx 2.02$, $df = 27$, $p \approx 0.0269$, reject at $\alpha = 0.05$, and describe a Type II error; targets using z with unknown σ and describing the wrong error direction.

· APSTAT-FRQ12-004: Unit 5 primary; Unit 1 secondary; randomized two-factor laboratory experiment comparing ice melt for Liners A and B across assigned ambient temperatures 15°C – 35°C , with separate models $\hat{y}_A = 18 + 2.4x$ and $\hat{y}_B = 42 + 1.6x$; graph and intersect lines at (30, 90), compare predictions at 20°C and 35°C , explain differential temperature sensitivity, and reject an always-better claim while limiting population scope; targets choosing a model solely by smaller slope and confusing randomized treatment assignment with random population sampling.